

Customer Churn Prediction Using Machine Learning

Abstract

Customer churn prediction is one of the most important applications of Business Analytics. This project aims to predict whether a customer is likely to discontinue using a company's service based on historical customer data. Machine learning algorithms are used to identify churn patterns, helping businesses improve customer retention and profitability.

Introduction

Customer churn refers to customers who stop using a company's products or services. Retaining existing customers is more cost-effective than acquiring new ones. Therefore, predicting churn enables organizations to take preventive actions and improve customer satisfaction.

Problem Statement

Many companies lose customers without understanding the reasons behind their decisions. The objective of this project is to build a predictive model that identifies customers who are likely to churn, enabling businesses to take proactive retention measures.

Objectives

- Analyze customer behavior.
- Identify factors influencing churn.
- Build a machine learning prediction model.
- Evaluate model performance.
- Provide business recommendations to reduce churn.

Scope

Telecom companies, Banking sector, Insurance companies, E-commerce platforms, Subscription-based services.

Literature Review

Previous research has shown that customer demographics, contract type, monthly charges, and service usage significantly influence customer churn. Machine learning models such as Logistic Regression, Decision Trees, and Random Forest have demonstrated good performance.

Dataset

Customer ID, Gender, Age, Contract Type, Monthly Charges, Total Charges, Internet Service, Tenure, Payment Method, Churn.

Tools

Python, Pandas, NumPy, Matplotlib, Scikit-learn, Jupyter Notebook.

Data Preprocessing

Remove duplicate records, handle missing values, encode categorical variables, normalize numerical features, split the data into training and testing sets.

EDA

Analyze customer age distribution, monthly charges, contract type, churn percentage, payment methods, and visualize them using charts.

Feature Engineering

Create variables such as average monthly spending, customer lifetime, total services subscribed, and contract duration.

Algorithms

Logistic Regression, Decision Tree, Random Forest, Support Vector Machine.

Model Training

Train and optimize the selected model using training data.

Evaluation

Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix.

Results

Example performance: Accuracy 89%, Precision 87%, Recall 85%, ROC-AUC 91%.

Business Insights

Customers with month-to-month contracts and high monthly charges are more likely to churn.

Advantages

Improves retention, increases profitability, supports data-driven decisions, reduces marketing costs.

Limitations

Depends on data quality, requires periodic updates, performance varies across industries.

Conclusion

Customer churn prediction enables organizations to proactively retain customers and improve long-term business performance.

References

IBM Telco Customer Churn Dataset, Scikit-learn Documentation, Pandas Documentation, NumPy Documentation, Matplotlib Documentation.